HireX — Executive Summary & Project Handover Overview

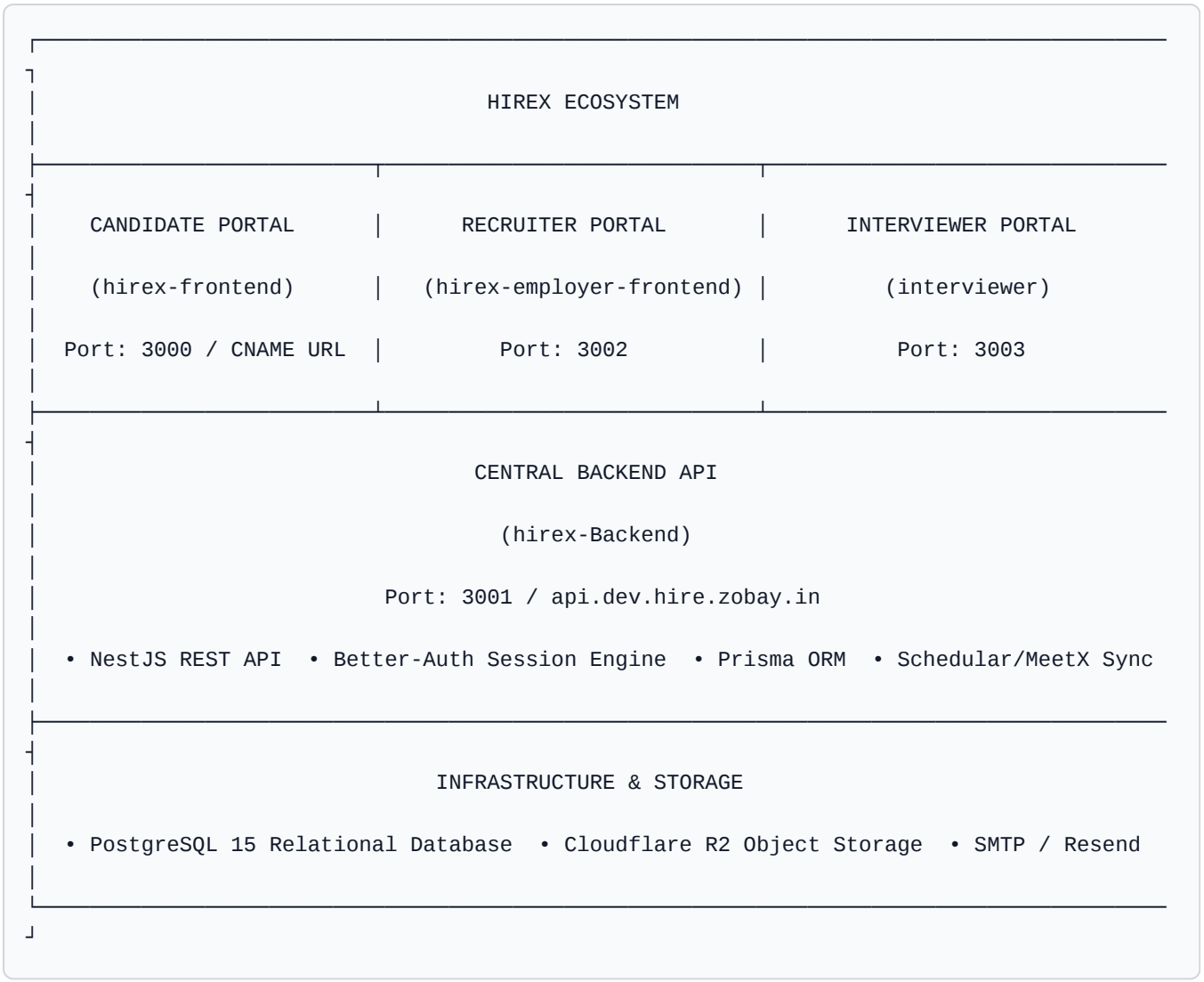
Document ID: `HXP-HANDOVER-00`
Platform Version: HireX V0 / Enterprise B2B Multi-Tenant Talent Operating System
Release Target: October 2026 Production Baseline
Audience: Software Architects, Engineering Leads, Product Managers, QA Leads, Security & DevOps Engineers

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1. Product Mission & Core Purpose

HireX is a unified, multi-tenant B2B talent acquisition platform that automates the end-to-end recruitment lifecycle from job requisition to post-offer compliance verification. The platform consolidates what typically requires four fragmented SaaS subscriptions:

1. **Applicant Tracking System (ATS)** with forward-only Kanban stage progression.
2. **Dynamic Application Form Builder** with custom per-job questions and candidate drafts.
3. **Automated Interview Scheduler (MeetX)** with recurring availability and WebRTC meeting links.
4. **Branded Offer & Document Verification Engine** with digital signatures, hourly expiration crons, and Cloudflare R2 multi-file verification.



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2. Key Modules & Functional Overview

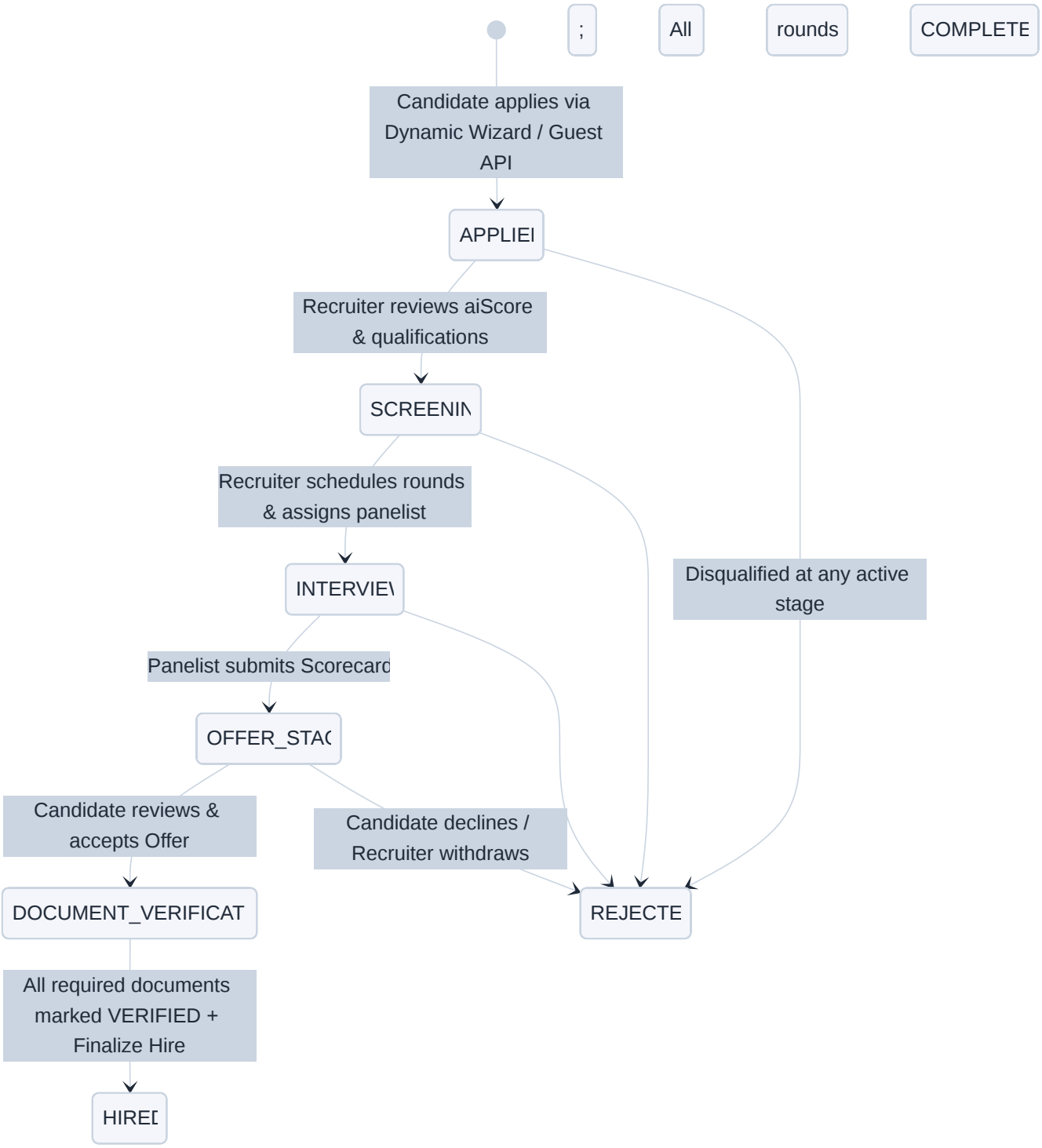
Module	Core Responsibility	Primary Technology	Key Entry Files
Authentication & Tenant Isolation	Multi-role cookie partitioning, invite tokens, tenant scoping by `companyId`	`better-auth`, NestJS Guards, Express cookies	`src/auth/`, `src/auth/session-cookie.ts`
Recruiter Dashboard	Job requisition, Kanban pipeline, dynamic form builder, offer letter generation, candidate analytics	Next.js App Router, Tailwind CSS, Recharts	`hirex-employer-frontend/src/app/`

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Candidate Portal	Job discovery, dynamic 3-step wizard with autosave, interview self-booking, offer review & e-sign	Next.js App Router, Custom Reverse Proxy	`hirex-frontend/src/app/`, `src/proxy.ts`
Interviewer Workspace	Assigned interview queue, weekly availability configuration, private recruiter briefings, rubric & cultural fit scorecards	Next.js App Router, Lucide Icons	`interviewer/src/app/`
Interview Orchestration & MeetX	Slot generation, atomic conflict resolution, Schedular (MeetX) video room lifecycle, webhook auto-completion	NestJS Services, Provider pattern, HMAC SHA-256	`src/interviews/`, `src/webhooks/`
Offer Letter Engine	Variable interpolation (`{{candidate_name}}` , `{{salary}}` , `{{expiry_date}}`), PDF generation, hourly expiration cron	`pdfkit` , Cloudflare R2 SDK, `@nestjs/schedule`	`src/offers/` , `src/recruiters/`
Document Verification	Multi-file document upload, in-browser preview, rejection with mandatory reason, candidate re-upload loop	Prisma `DocumentFile` , Cloudflare R2 signed URLs	`src/documents/`
Job Distribution & CNAME	Aggregator XML feeds (Indeed, LinkedIn, Naukri), custom CNAME domain reverse proxy mapping	`xmlbuilder2` , Next.js middleware / proxy	`src/recruiter/integrations/` , `src/proxy.ts`

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3. High-Level Recruitment Lifecycle (Forward-Only State Machine)

HireX strictly enforces a forward-only state machine to guarantee pipeline integrity:



Architectural Guardrails

1. **Reversion Block:** Moving backward from `SCREENING` or `INTERVIEW` to `APPLIED` is strictly rejected at the API and database levels.
2. **Open Interview Block:** Moving to `OFFER_STAGE` is blocked if any interview rounds are in `ASSIGNED` or `CONFIRMED` status.
3. **Mandatory Compliance Gate:** Candidates cannot move directly from `OFFER_STAGE` to `HIRED`. They must transition through `DOCUMENT_VERIFICATION`.

4. **Finalize Hire Guard:** The `HIRED` stage dropdown is locked; moving to `HIRED` requires clicking **"Finalize Hire"**, which enforces that all mandatory tenant documents are in `VERIFIED` status.

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4. Current Verification Baseline & System Health

The repository has been verified with clean, automated test and build execution:

- **TypeScript Compilation:** All 4 projects (`hirex-Backend` , `hirex-employer-frontend` , `hirex-frontend` , `interviewer`) compile with **0 errors** via `npx tsc --noEmit` .
- **Backend Unit & Integration Test Suite: 34 test suites passed (271 / 271 unit tests passed, 0 failures)**.
- **Hosted Staging Environments:** Active at `*.dev.hire.zobay.in` .

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5. Document Index & Reading Recommendations

Document File	Target Audience	Primary Focus
`01-TEST-PLAN.md`	QA Leads, Test Engineers	Test strategy, test levels, environments, entry/exit criteria
`02-TEST-CASES.md`	Manual & Automation QA	80+ granular test cases covering every page, role, endpoint, and edge case
`03-TEST-EXECUTION-REPORT.md`	QA, DevOps, Tech Leads	Recorded automated test run results, pass/fail counts, execution metrics
`04-PRD.md`	Product Managers, Business Analysts	Product requirements, user personas, acceptance criteria (Given/When/Then)
`05-TRD.md`	Senior Developers, Architects	System architecture, technical constraints, data contracts, technology choices
`06-ARCHITECTURE-AND-FLOWS.md`	All Developers	Sequence diagrams, data flow diagrams, state machines, component trees
`07-API-REFERENCE.md`	Backend & Integration Devs	Endpoint catalog (31 controllers), request/response schemas, error codes
`08-DATABASE-REFERENCE.md`	Backend Devs, DBAs	Schema reference, 25 Prisma models, 13 enums, indexes, cascade rules
`09-LOCAL-SETUP-AND-TROUBLESHOOTING.md`	Onboarding Engineers	Step-by-step setup for all 4 repos, seed commands, common error resolutions

Document File	Target Audience	Primary Focus
`10-SECURITY-AND-ACCESS-CONTROL.md`	Security Reviewers, Architects	Multi-tenant isolation (ADR-001), session security, webhook signatures
`11-RELEASE-AND-DEPLOYMENT-GUIDE.md`	DevOps & SRE Engineers	CI/CD build scripts, Docker configuration, DNS CNAME setup, migration steps
`12-KNOWN-ISSUES-AND-OPEN-QUESTIONS.md`	PMs & Tech Leads	Unresolved business decisions, missing vendor keys, technical debt register
`13-REQUIREMENT-TRACEABILITY.md`	QA Leads & PMs	Traceability matrix: Functional Req (FR) -> Component -> Test Cases -> Status
`14-REPOSITORY-INVENTORY.md`	New Developers	File tree inventory, directory purposes, configuration manifests
`README.md`	All Readers	Master index, orientation guide, and completion checklist